

Discriminants of the Pencils $H(W) - sW + t$

Repository working draft

July 21, 2026

Abstract

Let k be an algebraically closed field of characteristic zero and let $H \in k[W]$ have degree $n \geq 3$. The reduced discriminant of the pencil $H(W) - sW + t$ is parameterized by

$$r \longmapsto (H'(r), rH'(r) - H(r)).$$

We prove that for H in a nonempty Zariski-open set its projective closure is a rational plane curve of degree n , smooth at its unique point at infinity, with exactly $n - 2$ ordinary cusps and $\frac{1}{2}(n - 2)(n - 3)$ ordinary nodes, and with no other singularities. The same open set meets a natural normalized endpoint slice in every degree. The proof uses contact factorizations and dimension estimates; the genus formula is used only after all possible singularity types have been excluded.

1 Introduction

Pencils of the form

$$E_{s,t}(W) = H(W) - sW + t \tag{1.1}$$

occur when a polynomial inverse problem is reduced to choosing a marked root. In the motivating Jacobian-map construction, the target coordinates specialize to $s = BC$ and $t = cAC^2$. None of that construction is needed below: our subject is the intrinsic discriminant geometry of (1.1).

A repeated root r of (1.1) satisfies

$$s = H'(r), \quad t = rH'(r) - H(r). \tag{1.2}$$

Thus the discriminant records tangent lines to the affine graph $Y = H(W)$. Its cusps correspond to flexes and its self-intersections to bitangent lines. Rather than invoke a general theorem on generic dual curves, we use the special form (1.2) to control all bad contacts, including their behavior at infinity.

Throughout, “the discriminant curve” means the reduced projective closure of the repeated-root locus. Adding an affine-linear polynomial to H only applies an affine transformation to (s, t) , so we work in

$$\mathcal{B}_n = \operatorname{Spec} k[h_2, \dots, h_n], \quad H(W) = \sum_{j=2}^n h_j W^j, \tag{1.3}$$

and write $\mathcal{B}_n^\circ = D(h_n)$ for the exact-degree locus.

2 Statement of the theorem

Theorem 2.1. *For every $n \geq 3$ there is a nonempty Zariski-open subset $\mathcal{U}_n \subset \mathcal{B}_n^\circ$ such that for every $H \in \mathcal{U}_n$ the discriminant curve of $H(W) - sW + t$ has:*

1. *degree n and normalization $\mathbb{P}^1$;*
2. *one point at infinity, which is smooth;*
3. *exactly $n - 2$ ordinary cusps;*
4. *exactly $\frac{1}{2}(n - 2)(n - 3)$ ordinary nodes;*
5. *no other singularities.*

Moreover, $\mathcal{U}_n$ meets the normalized endpoint slice

$$\mathcal{A}_n = \{H : \deg H = n, H(0) = H'(0) = H(1) = 0, H'(1) = -1, H''(1) \neq -2\}. \quad (2.1)$$

Consequently the same conclusions hold on a nonempty open subset of $\mathcal{A}_n$.

The characteristic-zero assumption is used for derivatives, separability, and the local classification of the parameterized branches. The theorem is not asserted in small positive characteristic.

3 Normalization, degree, and infinity

Homogenizing (1.2) gives a morphism

$$\bar{\nu}_H : \mathbb{P}^1 \longrightarrow \mathbb{P}^2, \quad [R : U] \longmapsto [S_H : T_H : Z], \quad (3.1)$$

where

$$\begin{aligned} S_H &= \sum_{j=2}^n j h_j R^{j-1} U^{n-j+1}, \\ T_H &= \sum_{j=2}^n (j-1) h_j R^j U^{n-j}, \\ Z &= U^n. \end{aligned}$$

These degree- n forms have no common zero on $D(h_n)$. The only parameter above the line at infinity is $[1 : 0]$, and its image is $[0 : 1 : 0]$.

Proposition 3.1. *For every $H \in \mathcal{B}_n^\circ$, the map $\bar{\nu}_H$ is finite and birational onto an integral plane curve of degree n . It is a formal closed immersion at $[1 : 0]$; in particular its image is smooth there and no off-diagonal equal-image pair can specialize to that point while h_n remains nonzero.*

Proof. On the affine chart,

$$\frac{ds}{dr} = H''(r), \quad \frac{dt}{dr} = rH''(r).$$

At the generic point of the image, $r = dt/ds$; hence the map is birational. A nonconstant map from the proper curve $\mathbb{P}^1$ to its image is finite. The base-point-free forms in (3.1) pull a general line back to a divisor of degree n . Birationality therefore makes the image degree n .

For the assertion at infinity, put $q = U/R$ and use the chart $T_H \neq 0$. Uniformly over $k[h_2, \dots, h_n, h_n^{-1}]$,

$$x := \frac{S_H}{T_H} = \frac{n}{n-1}q + O(q^2), \quad y := \frac{Z}{T_H} = \frac{q^n}{(n-1)h_n} + O(q^{n+1}). \quad (3.2)$$

The first series has unit linear coefficient, so the formal inverse theorem recovers q uniquely from x . The completed image is therefore the graph $y = f(x)$, proving formal closed immersion and smoothness.

For completeness, consider the relative fiber product of two copies of $\mathbb{P}^1$ over the image while the coefficients vary in $D(h_n)$. If an off-diagonal pair approached the infinity section, the uniqueness of the preimage of $Z = 0$ forces both parameters to approach $q = 0$. In the completed neighborhood, equality of the two x -coordinates and formal invertibility in (3.2) force $q_1 = q_2$. Thus the off-diagonal locus has closure disjoint from the infinity section. $\square$

4 Contact strata and exclusion of bad singularities

Let ℓ_r denote the tangent line to $Y = H(W)$ at r , and set

$$c(r) = \text{ord}_{W=r}(H(W) - \ell_r(W)) \geq 2.$$

Differentiation of (1.2) gives

$$\nu'_H(r) = H''(r)(1, r). \quad (4.1)$$

Thus $c(r) = 2$ gives a smooth immersed branch. If $c(r) = 3$, then $H'''(r) \neq 0$ and

$$\det(\nu''_H(r), \nu'''_H(r)) = 2H'''(r)^2 \neq 0, \quad (4.2)$$

so the image is an ordinary cusp. Contact $c(r) \geq 4$ is the first bad one-branch event.

If two distinct parameters r, u have the same image, the graph has a common tangent line at both points. When both branches are unramified their tangent directions are $(1, r)$ and $(1, u)$, with determinant $u - r \neq 0$. The intersection is automatically an ordinary transverse node. The same determinant is the Jacobian of the equal-image equations with respect to the two parameters, so the tangent-chord coincidence is transverse as an incidence as well. In particular a tacnode cannot occur from two distinct unramified normalization points.

It remains to show that all other events are nongeneric. For a contact pattern $\mu = (\mu_1, \dots, \mu_k)$, write

$$M_\mu(W) = \prod_{i=1}^k (W - r_i)^{\mu_i}, \quad m = \sum_i \mu_i.$$

A line with these contacts gives the factorization

$$H(W) - \ell(W) = M_\mu(W)Q(W), \quad \deg Q \leq n - m. \quad (4.3)$$

After discarding the affine-linear coefficients, (4.3) defines a polynomial map to $\mathcal{B}_n$ from a space of dimension

$$k + (n - m + 1). \quad (4.4)$$

For each of the three patterns

$$(4), \quad (3, 2), \quad (2, 2, 2), \quad (4.5)$$

this dimension equals $n - 2$ whenever the pattern is possible. The Zariski closure of the image of a finite-type morphism cannot have larger dimension than its source. Hence the closed image $Z_\mu \subset \mathcal{B}_n$ of every pattern in (4.5) has codimension at least one. This argument already takes the closure: collisions among marked roots, roots of Q meeting a marked root, and degree drops in Q cannot create a larger boundary component.

Lemma 4.1. *Outside $Z_{(4)} \cup Z_{(3,2)} \cup Z_{(2,2,2)}$, every affine singularity of the discriminant curve is an ordinary node or ordinary cusp.*

Proof. A nonordinary single branch has contact at least four and is in $Z_{(4)}$. If a fiber of the normalization contains two points and one branch is ramified, the contact pattern dominates $(3, 2)$. If it contains at least three points, any three give $(2, 2, 2)$. These statements include higher contact, cusp–cusp collisions, tritangents, and two nominal node pairs with the same image. The only remaining case has exactly two unramified preimages, and (4.1) makes their branches transverse. $\square$

The proposition at infinity is essential here: it prevents a missing projective bad stratum. A ramification point cannot move to parameter infinity on $D(h_n)$ because the homogenization of H'' there has value $n(n - 1)h_n$. Nor can a finite branch share the image $[0 : 1 : 0]$. Finally, the relative fiber-product statement in Proposition 3.1 rules out two finite equal-image parameters escaping together to infinity. Thus (4.5) exhausts the projective, not merely affine, failure modes.

The complement

$$\mathcal{U}_n = \mathcal{B}_n^\circ \setminus (Z_{(4)} \cup Z_{(3,2)} \cup Z_{(2,2,2)}) \quad (4.6)$$

is nonempty and open. On it, H'' is squarefree, so its $n - 2$ roots give $n - 2$ distinct ordinary cusps. No cusp shares its image with another branch, and every other singularity is an ordinary node.

5 The endpoint slice is not trapped in the bad locus

We now prove the last genericity assertion in Theorem 2.1. It is not enough to know that (4.6) is nonempty in the full coefficient space.

For $G \in \mathcal{B}_n$ and $\alpha \neq \beta$, define the divided tangent–chord equation

$$\Phi(G, \alpha, \beta) = \frac{G(\beta) - G(\alpha) - G'(\alpha)(\beta - \alpha)}{(\beta - \alpha)^2} = 0. \quad (5.1)$$

It is polynomial, and the coefficient of h_2 in Φ is one. Therefore the incidence $\mathcal{T}_n = V(\Phi)$ over $\alpha \neq \beta$ is an irreducible affine bundle. Its projection to $\mathcal{B}_n^\circ$ is surjective: for any exact-degree G , choose α with $G''(\alpha) \neq 0$. As a polynomial in β , (5.1) has degree $n - 2$ and does not vanish at $\beta = \alpha$, so it has a root distinct from α .

Put $d = \beta - \alpha$ and

$$H(W) = G(\alpha + dW) - G(\alpha) - G'(\alpha)dW. \quad (5.2)$$

Equation (5.1) gives $H(0) = H'(0) = H(1) = 0$. Source reparameterization and an invertible affine transformation of (s, t) carry the discriminant of G to that of H , preserving all projective singularity types, including the point at infinity. Hence the inverse image of $\mathcal{U}_n$ in $\mathcal{T}_n$ is a nonempty open.

The additional conditions

$$\deg H = n, \quad H'(1) \neq 0, \quad H''(1) - 2H'(1) \neq 0 \quad (5.3)$$

also define a nonempty open in $\mathcal{T}_n$. A uniform witness is

$$H_n(W) = W^2 + W^3 + \cdots + W^{n-1} - (n - 2)W^n, \quad (\alpha, \beta) = (0, 1), \quad (5.4)$$

for which $H_n(1) = 0$, $H'_n(1) \neq 0$, and $H''_n(1)/(-H'_n(1)) = -4n/3 \neq -2$. Since $\mathcal{T}_n$ is irreducible, the two nonempty opens intersect. Scaling (5.2) by $-1/H'_n(1)$ reaches $H'_n(1) = -1$ without changing any singularity type or the last inequality in (5.3). This proves that $\mathcal{A}_n \cap \mathcal{U}_n$ is nonempty.

6 Counting the nodes

Only now do we use the genus formula. For $H \in \mathcal{U}_n$, the curve is integral of degree n with normalization $\mathbb{P}^1$. Sections 3 and 4 have already excluded every singularity except ordinary nodes and the $n - 2$ ordinary cusps. Both types have delta invariant one. If N denotes the number of nodes, then

$$\frac{(n-1)(n-2)}{2} = p_a(C) = g(\mathbb{P}^1) + \sum_{x \in \text{Sing } C} \delta_x = 0 + (n-2) + N.$$

Thus

$$N = \frac{(n-2)(n-3)}{2}.$$

The genus formula counts the nodes left after the geometric exclusion; it is not used to infer that unclassified singularities must be nodes.

References

The proof uses only standard facts about finite maps of proper curves, dimension of images, formal inversion, and the normalization-genus formula. For proper images see the Stacks Project, Tag 01W6. For broader context on inflections and bitangents as singularities of dual curves, see A. Yuran, *Plücker formulas for plane algebraic curves with a given Newton polygon*, arXiv:2204.04626.